Empathetic Dialogues Classification

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What is empathy?

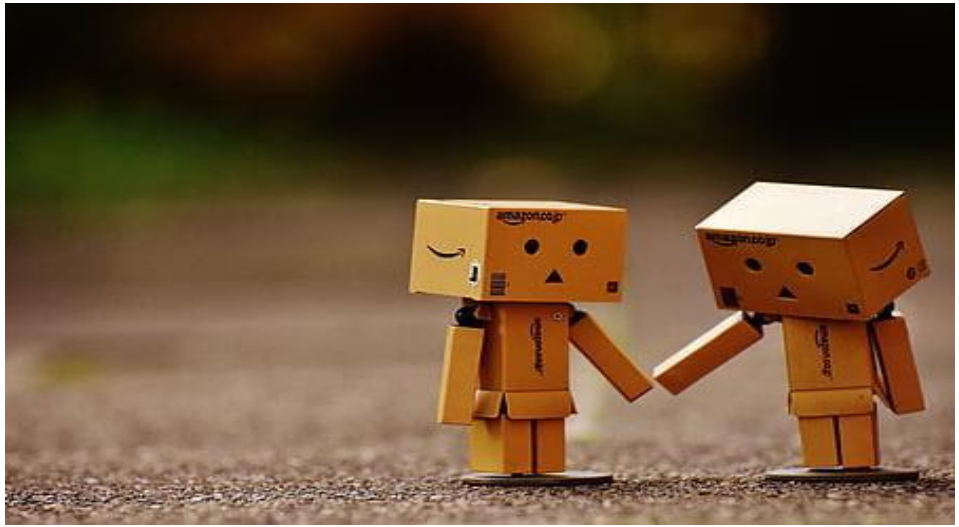
Empathy is the ability to understand and share the feelings of others.

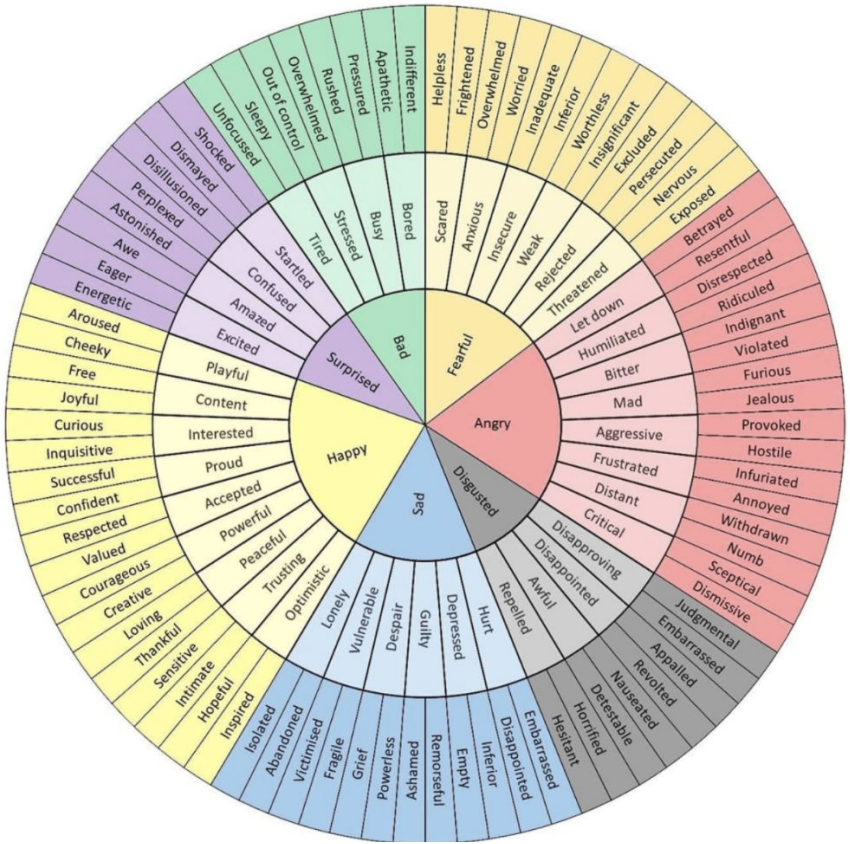
Problem:

Comparing traditional ML and deep learning techniques to build a text-based emotion/empathy classifier.

Goal:

To evaluate the effectiveness traditional ML vs. deep learning in emotion analysis





Advancements in NLP:

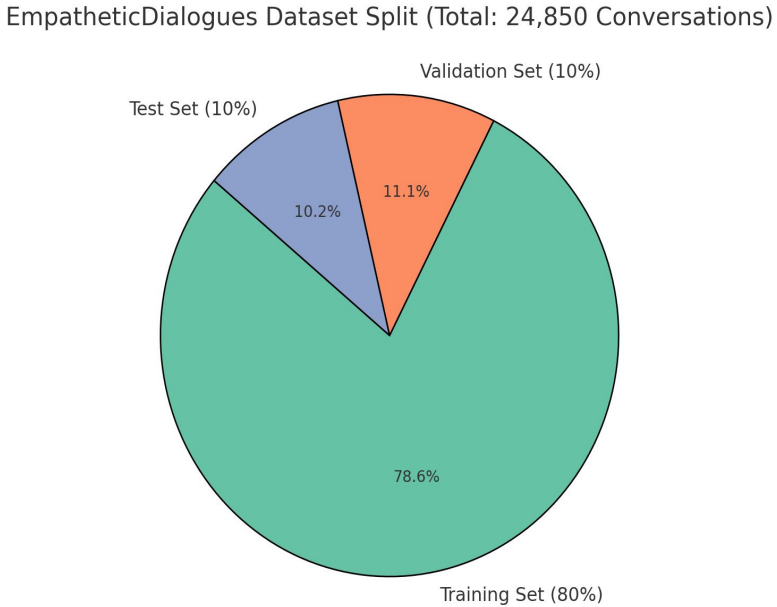
- Models like BERT excel at understanding but require fine-tuning to effectively capture emotional context.
- *EmpatheticDialogues* Dataset: 25,000 empathetic conversations to train AI on recognizing emotion in text.

Goal:

Develop a machine learning and a deep learning model to classify the emotion of conversations in *EmpatheticDialogues.*

Dataset

- *EmpatheticDialogues* is a dataset designed to train models to recognize, understand, and respond appropriately to human emotions in conversations.
- Dataset Overview:
  - Total conversations: 24,850
  - Total Participants: 810 (Amazon MTurk)
- Dataset Split:
  - Training set: 19,533 conversations (~78%)
  - Validation set: 2,770 conversations (~11%)
  - Test set: 2,547 conversations (~10%)



Dataset and Labels

Each conversation has a Speaker (who shares an emotional situation) and a Listener (who responds empathetically).

Label: Afraid
Situation: Speaker felt this when...
"I've been hearing noises around the house at night"
Conversation:
Speaker: I've been hearing some strange noises around the house at night.
Listener: oh no! That's scary! What do you think it is?
Speaker: I don't know, that's what's making me anxious.
Listener: I'm sorry to hear that. I wish I could help you figure it out

Label: Proud
Situation: Speaker felt this when...
"I finally got that promotion at work! I have tried so hard for so long to get it!"
Conversation:
Speaker: I finally got promoted today at work!
Listener: Congrats! That's great!
Speaker: Thank you! I've been trying to get it for a while now!
Listener: That is quite an accomplishment and you should be proud!

Dataset and Labels

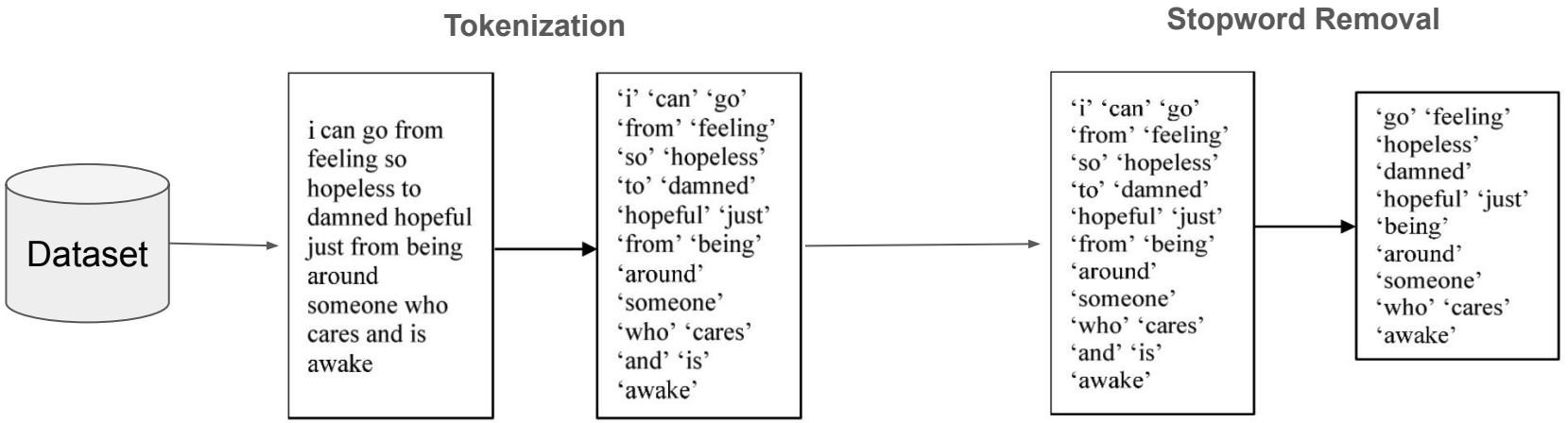
- Includes 32 emotion labels.
- Displays frequency of different emotions in dataset.
- Highlights most-used words by **Speaker** and **Listener** for each emotion.

Emotion	Most-used speaker words	Most-used listener words	Training set emotion distrib
Surprised	got,shocked,really	that's,good,nice	5.1%
Excited	going,wait,i'm	that's,fun,like	3.8%
Angry	mad,someone,got	oh,would,that's	3.6%
Proud	got,happy,really	that's,great,good	3.5%
Sad	really,away,get	sorry,oh,hear	3.4%
Annoyed	get,work,really	that's,oh,get	3.4%
Grateful	really,thankful,i'm	that's,good,nice	3.3%
Lonely	alone,friends,i'm	i'm,sorry,that's	3.3%
Afraid	scared,i'm,night	oh,scary,that's	3.2%
Terrified	scared,night,i'm	oh,that's,would	3.2%
Guilty	bad,feel,felt	oh,that's,feel	3.2%
Impressed	really,good,got	that's,good,like	3.2%
Disgusted	gross,really,saw	oh,that's,would	3.2%
Hopeful	i'm,get,really	hope,good,that's	3.2%
Confident	going,i'm,really	good,that's,great	3.2%
Furious	mad,car,someone	oh,that's,get	3.1%
Anxious	i'm,nervous,going	oh,good,hope	3.1%
Anticipating	wait,i'm,going	sounds,good,hope	3.1%
Joyful	happy,got,i'm	that's,good,great	3.1%
Nostalgic	old,back,really	good,like,time	3.1%
Disappointed	get,really,work	oh,that's,sorry	3.1%
Prepared	ready,i'm,going	good,that's,like	3%
Jealous	friend,got,get	get,that's,oh	3%
Content	i'm,life,happy	good,that's,great	2.9%
Devastated	got,really,sad	sorry,oh,hear	2.9%
Embarrassed	day,work,got	oh,that's,i'm	2.9%
Caring	care,really,taking	that's,good,nice	2.7%
Sentimental	old,really,time	that's,oh,like	2.7%
Trusting	friend,trust,know	good,that's,like	2.6%
Ashamed	feel,bad,felt	oh,that's,i'm	2.5%
Apprehensive	i'm,nervous,really	oh,good,well	2.4%
Faithful	i'm,would,years	good,that's,like	1.9%

Preprocessing Steps - Combining conversations

Combined Conversation (Speaker + Listener)	Label
“I've been hearing some strange noises around the house at night. Oh no! That's scary! What do you think it is? I don't know, that's what's making me anxious. I'm sorry to hear that. I wish I could help you figure it out.”	Afraid
“I finally got promoted today at work! Congrats! That’s great! Thank you! I’ve been trying to get it for a while now! That is quite an accomplishment and you should be proud!	Proud

Preprocessing Steps - Tokenization & Stopword Removal



Preprocessing Steps - TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) measures how important a word is in a document relative to a collection of documents. It helps to identify key terms by reducing the weight of common words.

$$TF = \frac{Number\ of\ times\ a\ word\ "X"\ appears\ in\ a\ Document}{Number\ of\ words\ present\ in\ a\ Document}$$

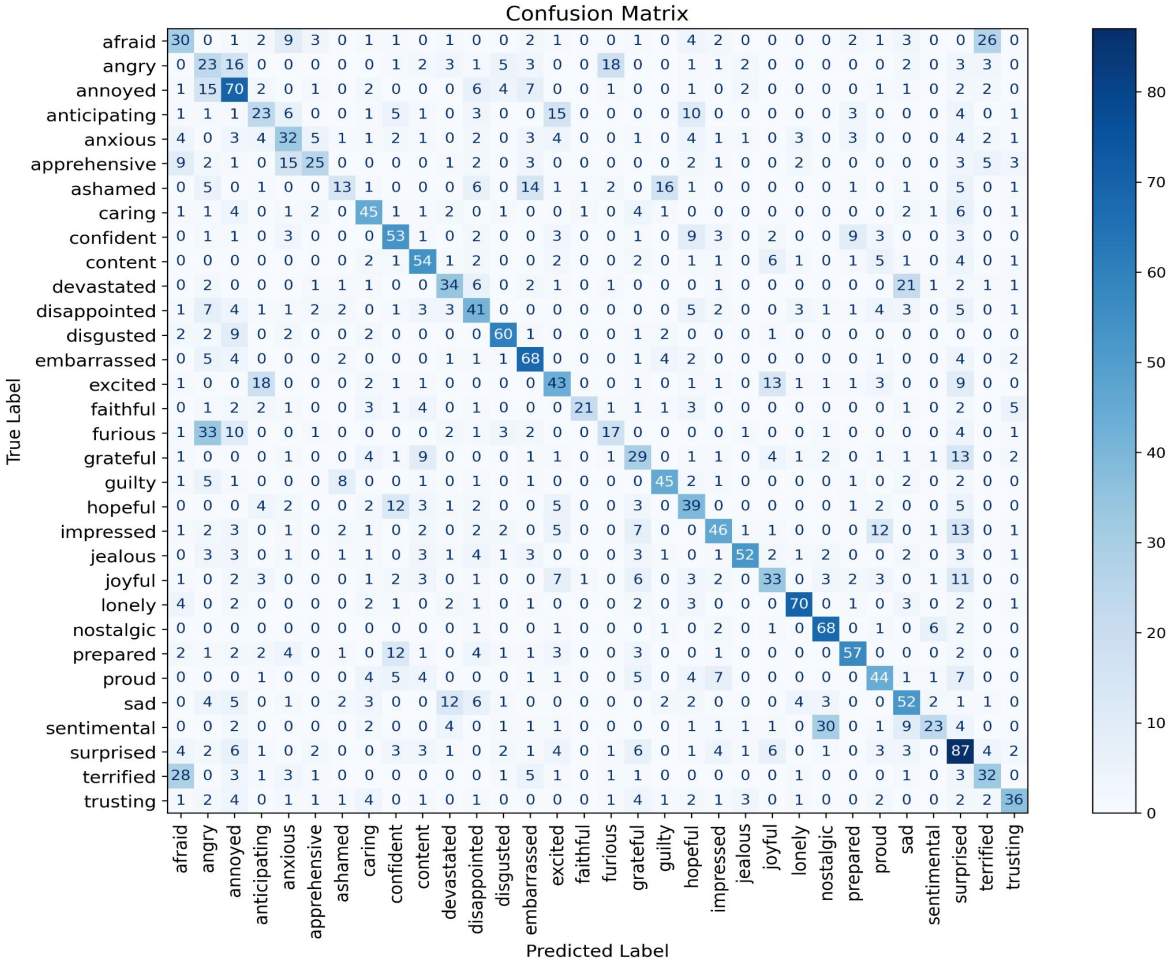
$$IDF = log\left(\frac{Number\ of\ Documents\ present\ in\ a\ Corpus}{Number\ of\ Documents\ where\ word\ "X"\ has\ appeared}\right)$$

$$TF\ IDF\ =\ TF * IDF$$

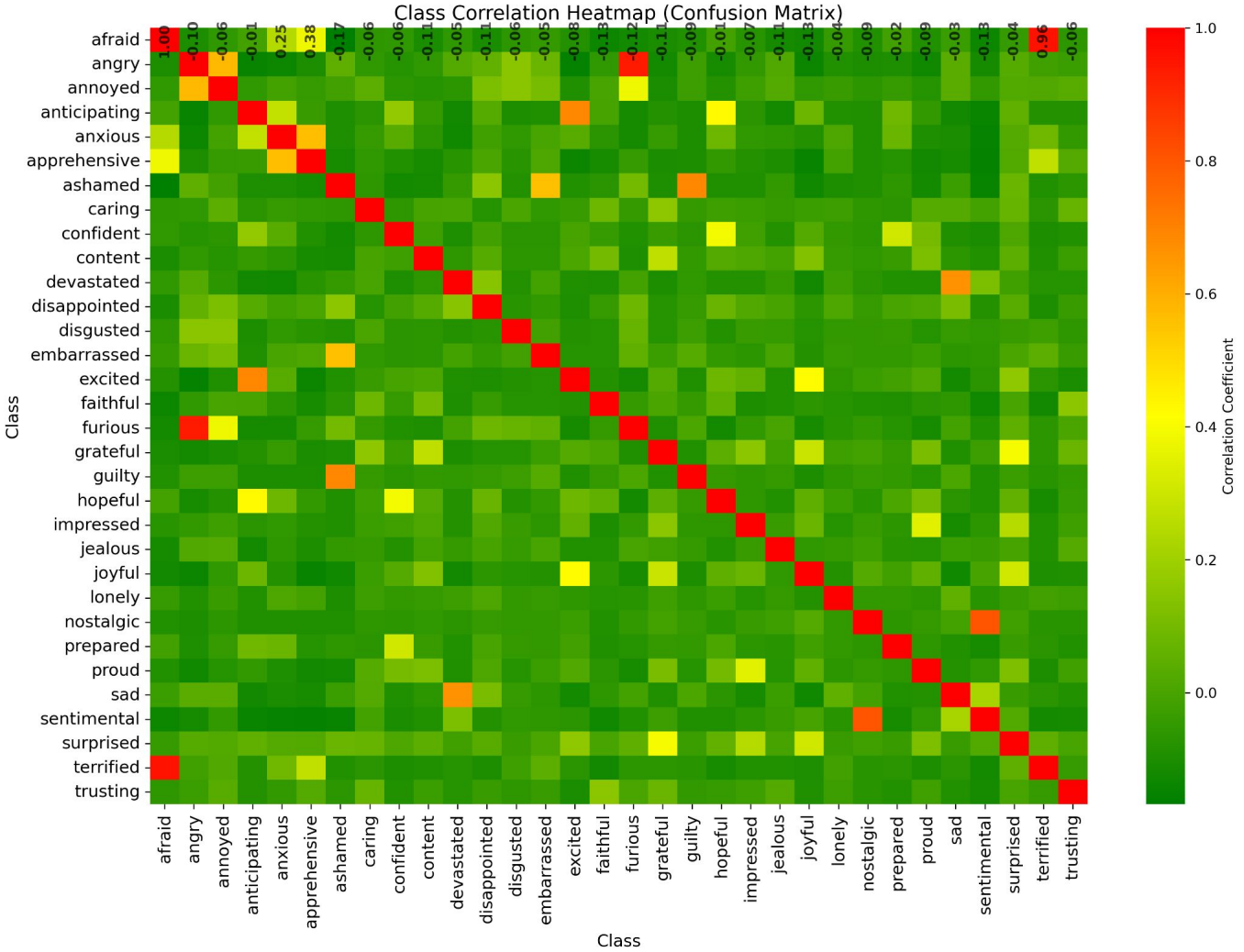
Classification using Traditional Models

Model	Training error	Validation Error
Linear SVM	0.7986	0.8125
Logistic regression	0.3222	0.5548
Random Forest	0.3868	0.6609
Naive Bayes	0.1125	0.5802
Decision Tree	0.0	0.7505

Confusion Matrix using SVM - Validation Set



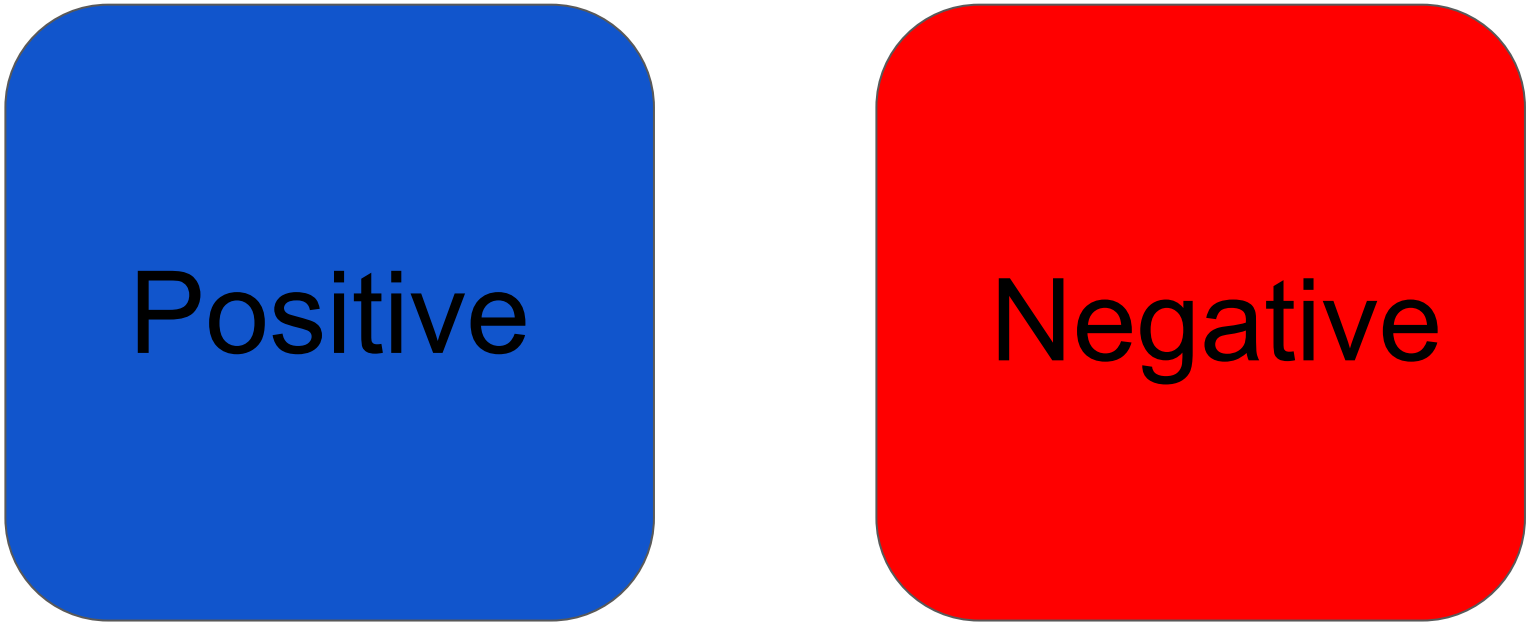
Class Correlation Heat Map



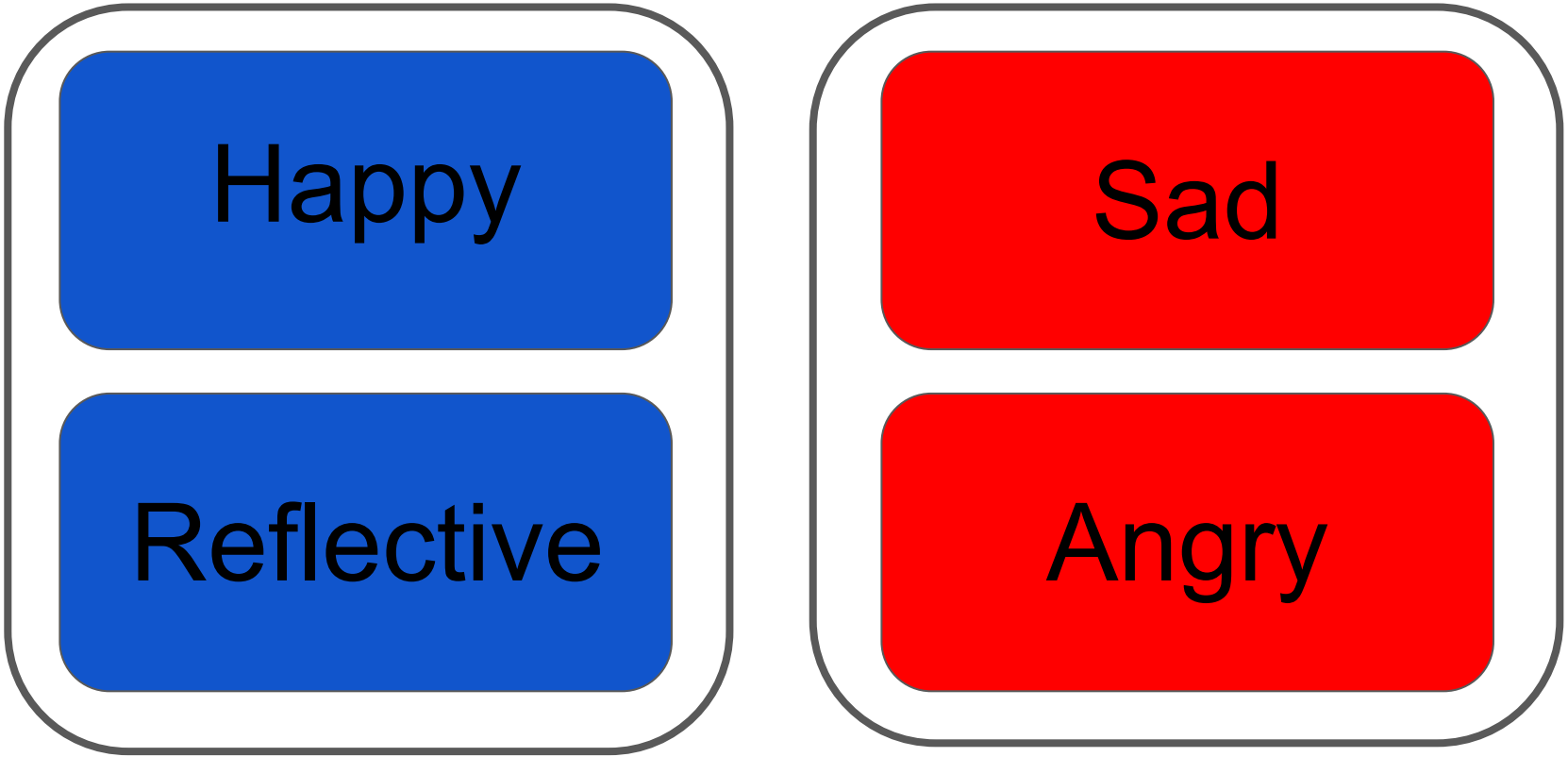
Emotions Mapping Scheme

Emotion	Relabelled
afraid	fear
terrified	
angry	anger
annoyed	
furious	
anxious	worried
apprehensive	
ashamed	shame
embarrassed	
guilty	
hopeful	hope
confident	
prepared	
sad	sadness
devastated	
excited	happy
anticipating	
joyful	
impressed	recognition
proud	
nostalgic	nostalgia
sentimental	
jealous	jealous
grateful	grateful
surprised	surprised
caring	caring
lonely	lonely
faithful	faithful
disappointed	disappointed
disgusted	disgusted
content	content
trusting	trusting

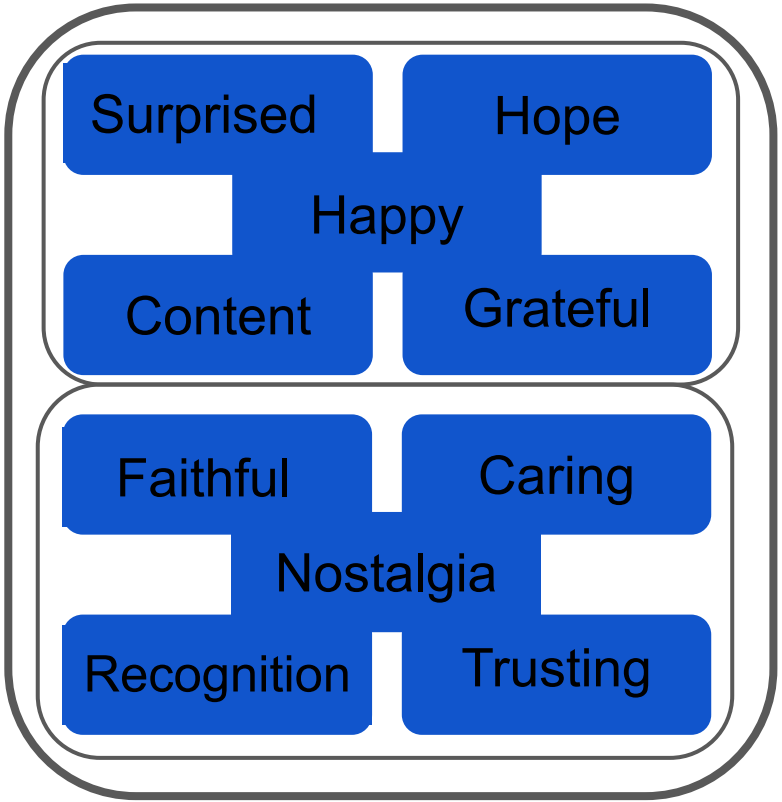
Hierarchical Model

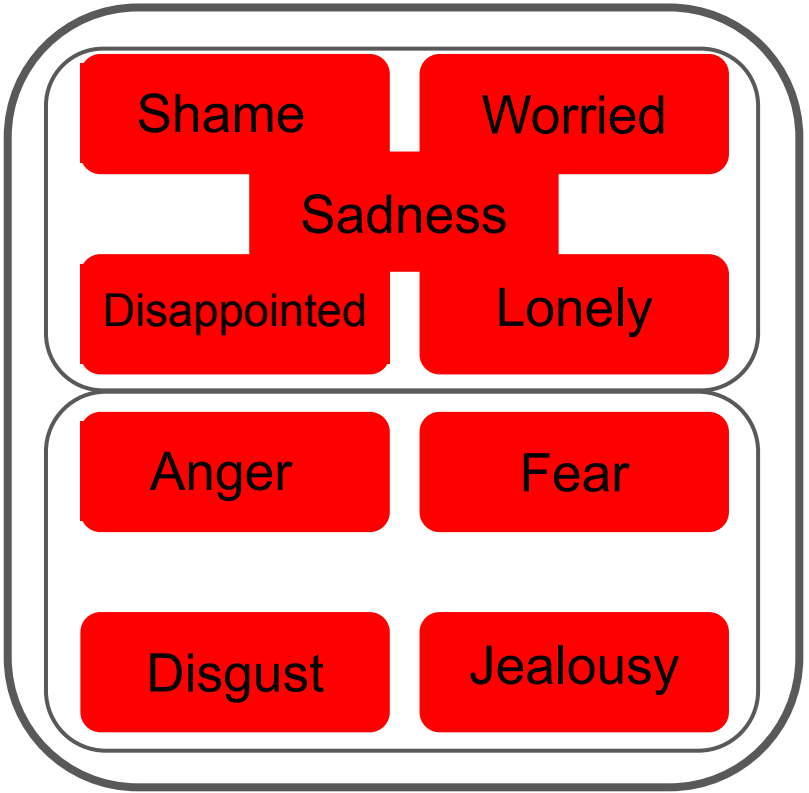


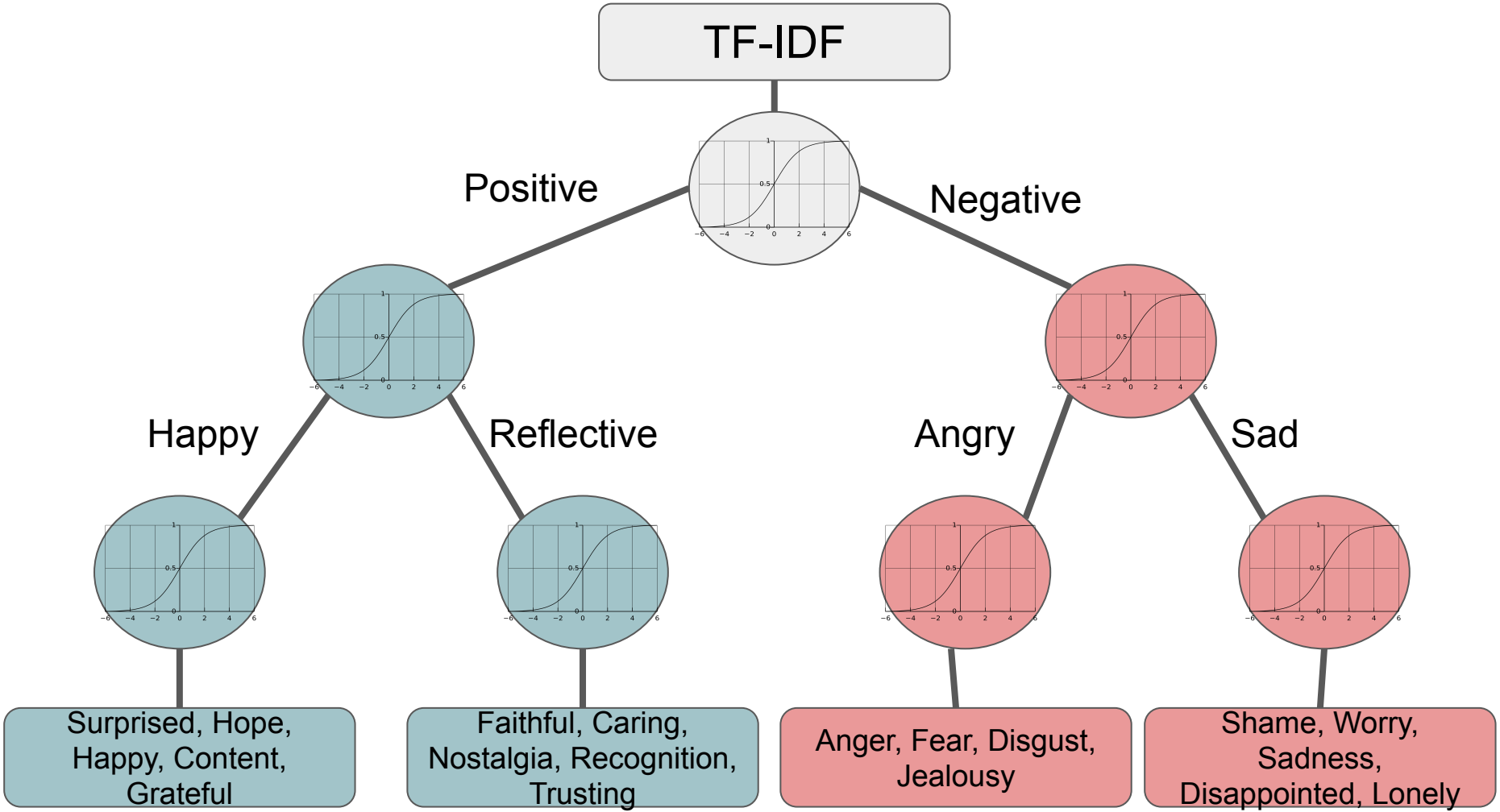
Hierarchical Model

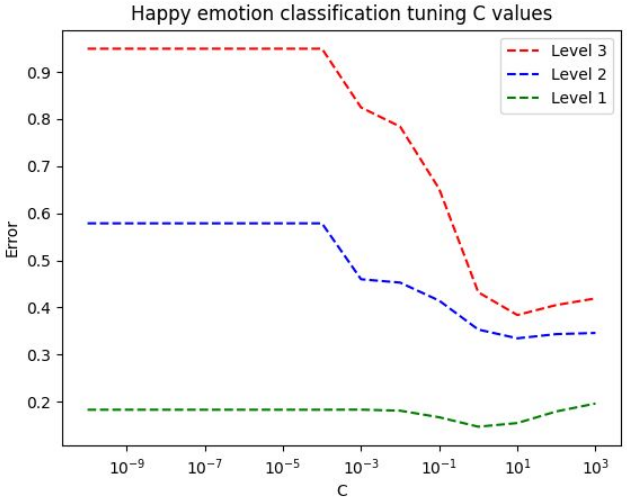


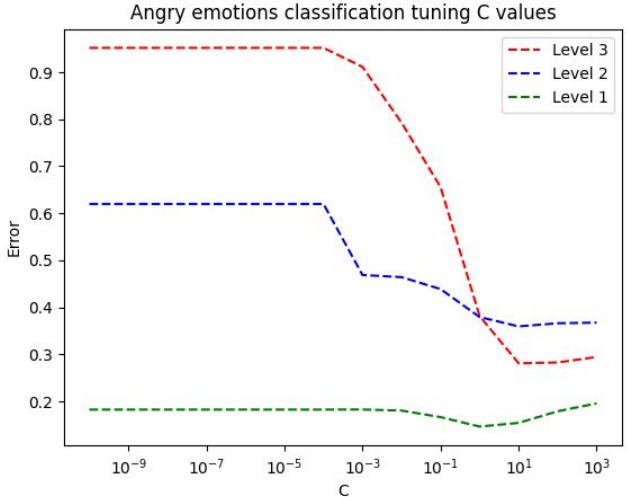
Hierarchical Model

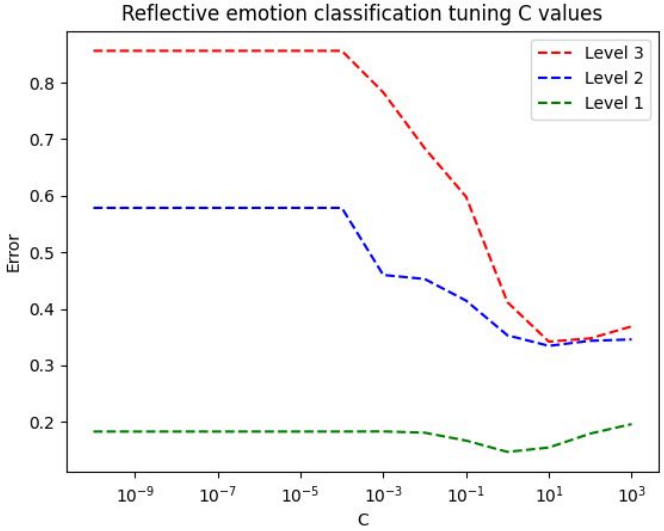


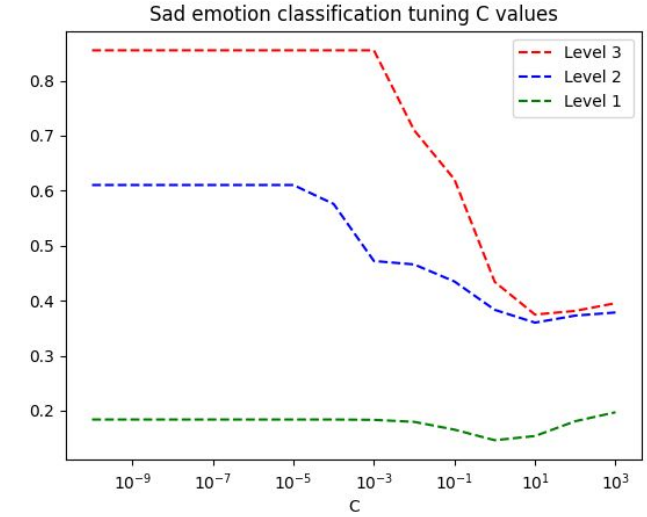



TF-IDF
Positive
Negative
Happy
Reflective
Angry
Sad
Surprised, Hope, Happy, Content, Grateful
Faithful, Caring, Nostalgia, Recognition, Trusting
Anger, Fear, Disgust, Jealousy
Shame, Worry, Sadness, Disappointed, Lonely



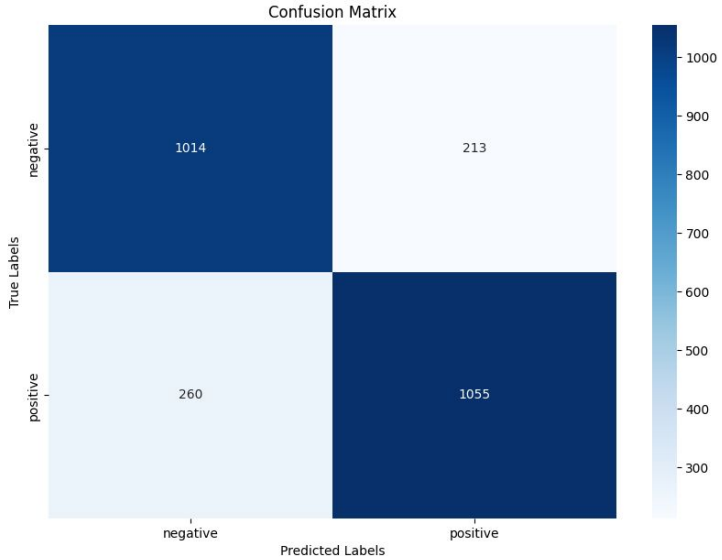


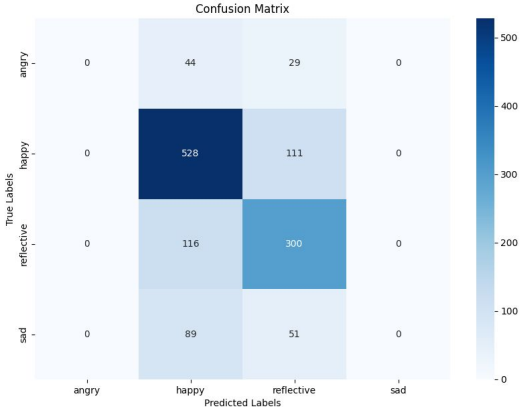


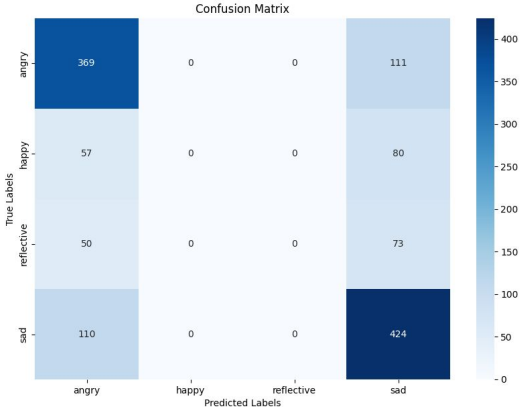


As classification becomes finer, the regularization strength decreases.

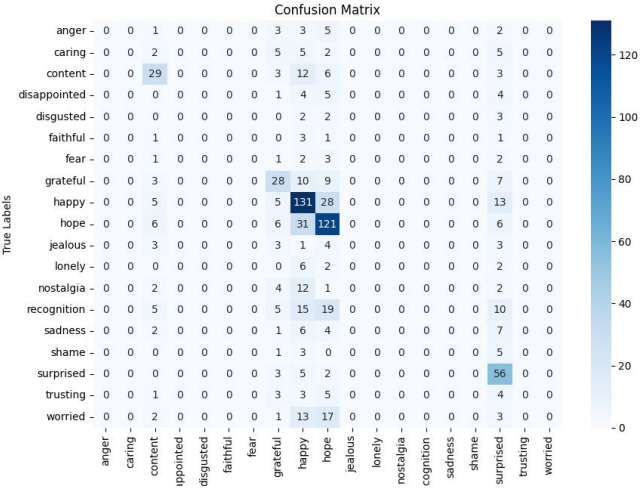
*In scikit-learn logistic regression, smaller C values induce stronger regularization.


Confusion Matrix
1014
213
260
1055
negative
positive
True Labels
Predicted Labels
1000
900
800
700
600
500
400
300

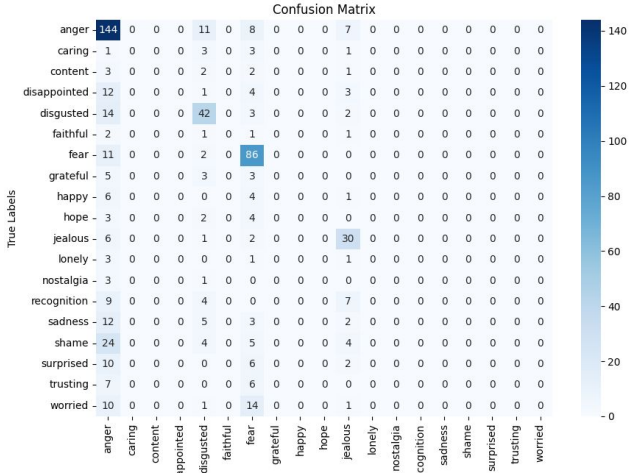

Confusion Matrix
0 44 29 0
0 528 111 0
0 116 300 0
0 89 51 0
angry
happy
reflective
sad
True Labels
Predicted Labels
500
400
300
200
100
0


Confusion Matrix
369 0 0 111
57 0 0 80
50 0 0 73
110 0 0 424
angry
happy
reflective
sad
True Labels
Predicted Labels
400
350
300
250
200
150
100
50
0

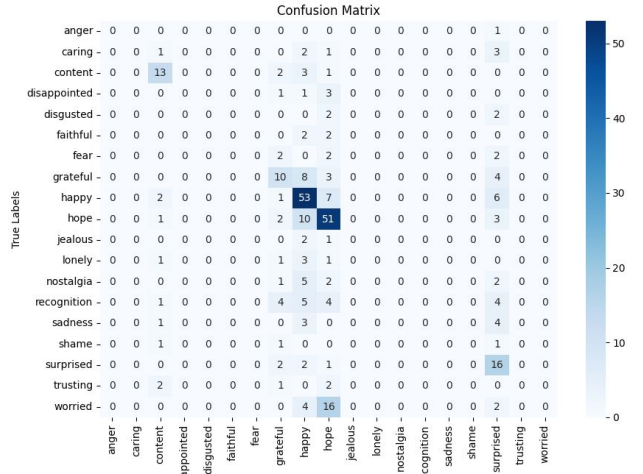
Happy Classification



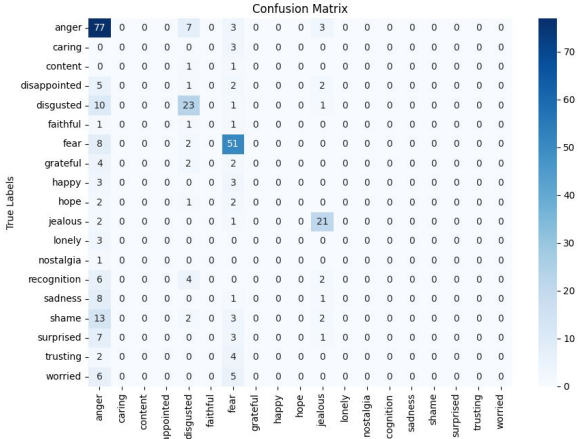
Angry Classification

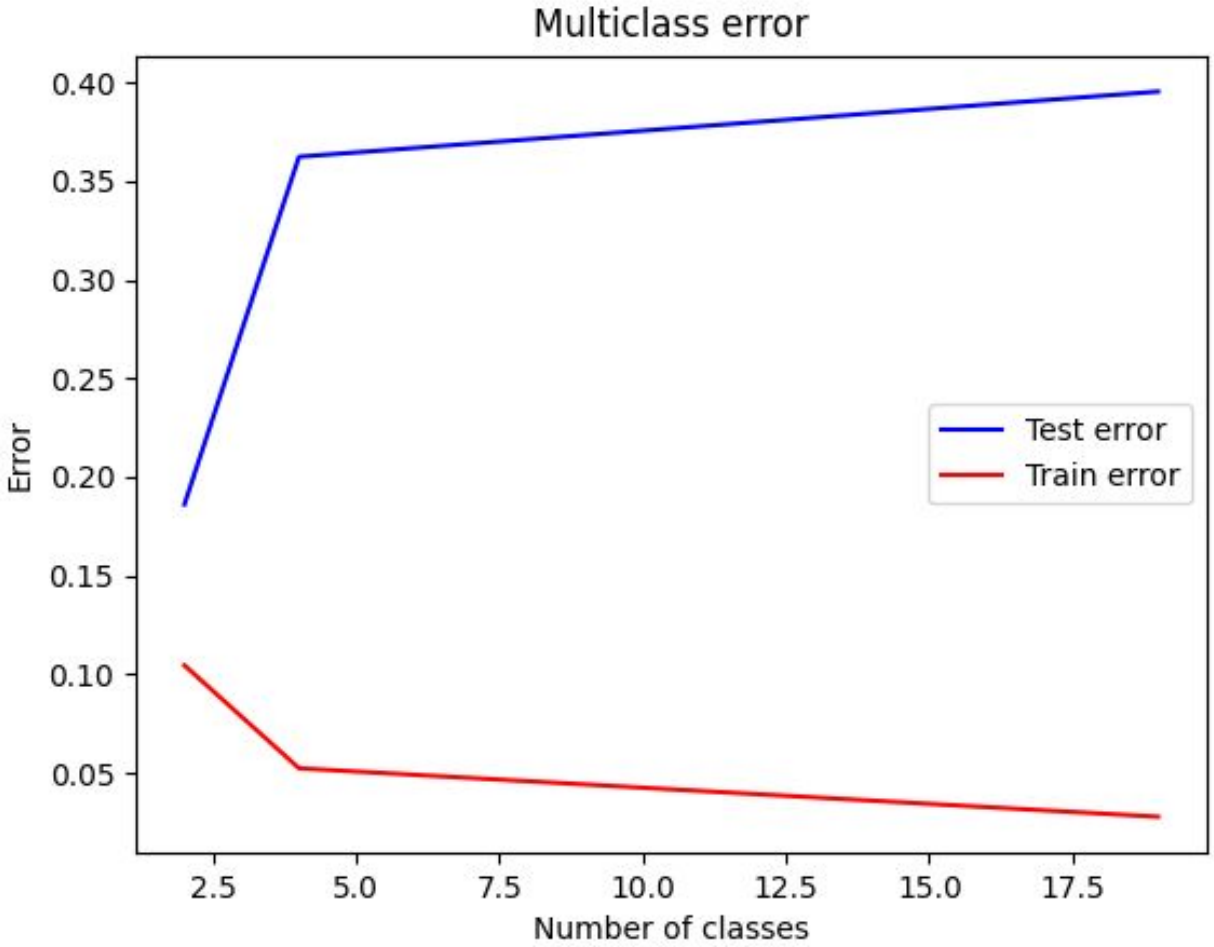


Reflective Classification



Sad Classification





Error increases as the number of classes increases.

Final Error:
0.3954 ± 0.0190

*Random classification would have an error of .9474

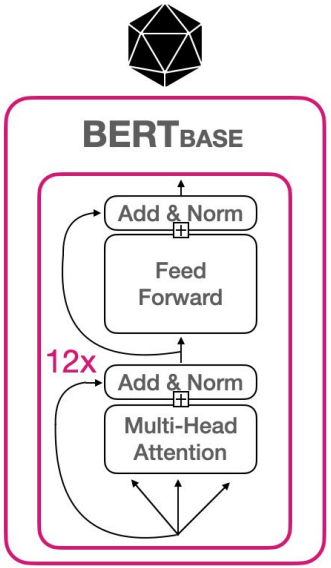
BERT - Bidirectional Encoder Representations from Transformers

- BERT is a deep learning model developed by Google that excels in **natural language understanding (NLU)** tasks like text classification, question answering, and text generation.

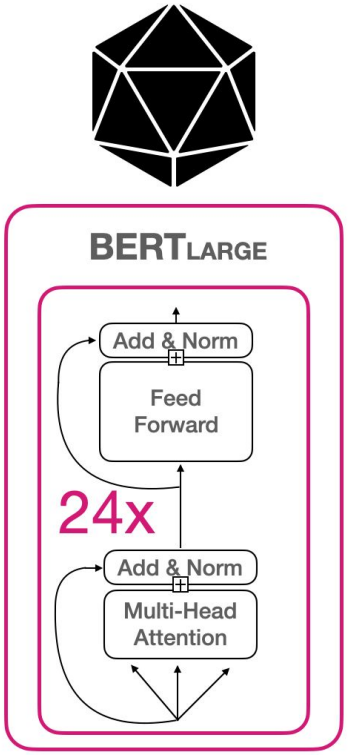
- Unlike previous models that process text from left to right (e.g., GPT) or right to left, BERT **reads the entire sentence at once**, learning context from both directions.

- **Pre-trained** on large datasets (like Wikipedia & BooksCorpus) and then **fine-tuned** on specific tasks like sentiment analysis or text classification.

BERT - Base vs Large LLM



110M Parameters

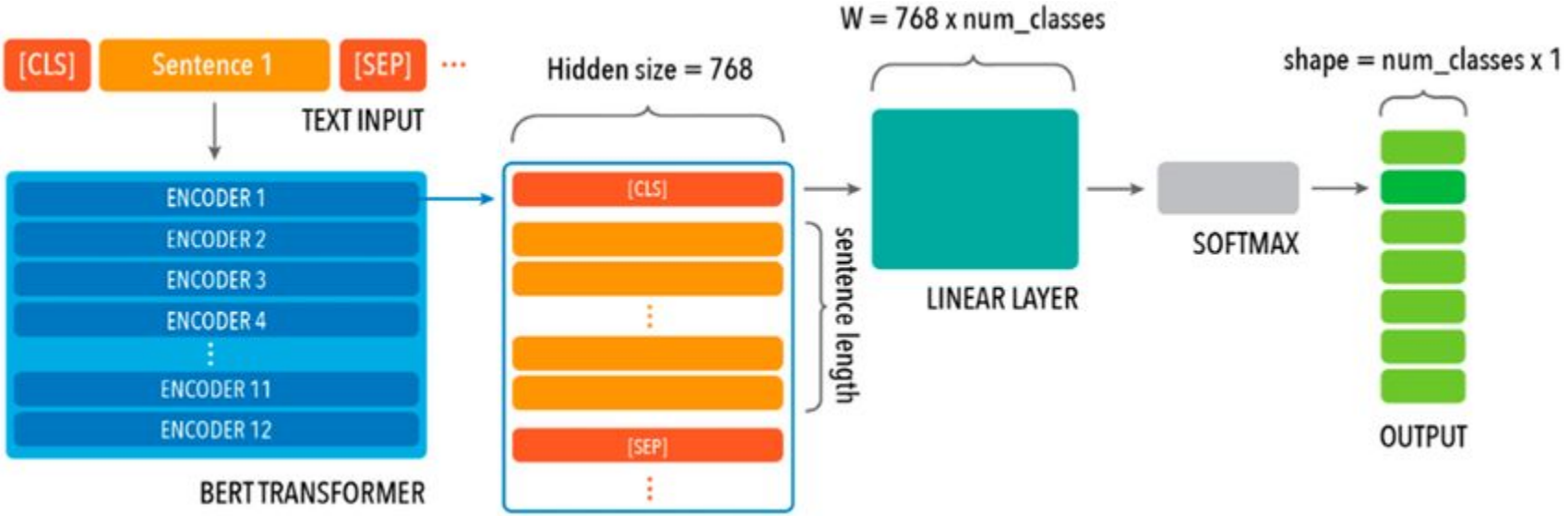


340M Parameters



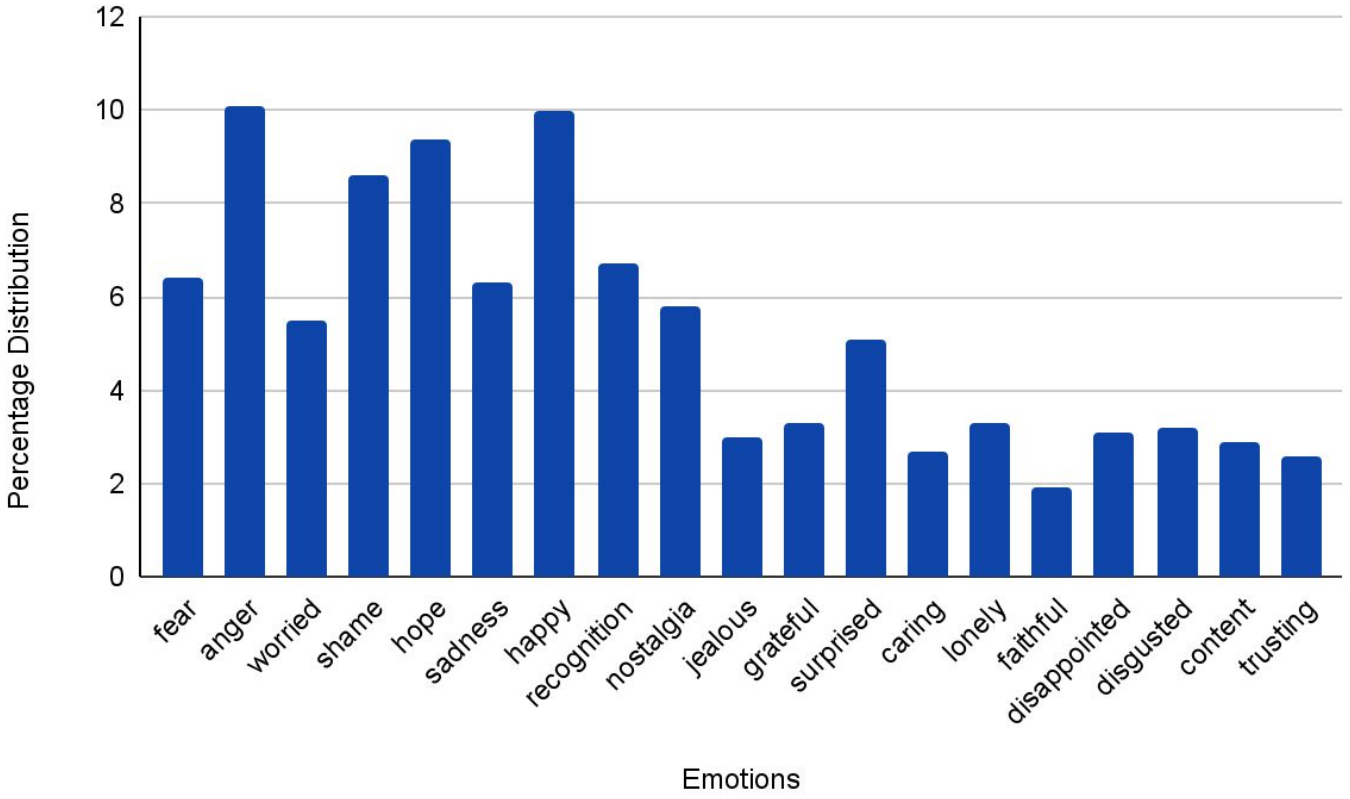
Source: https://huggingface.co/blog/bert-101

BERT Base - Architecture

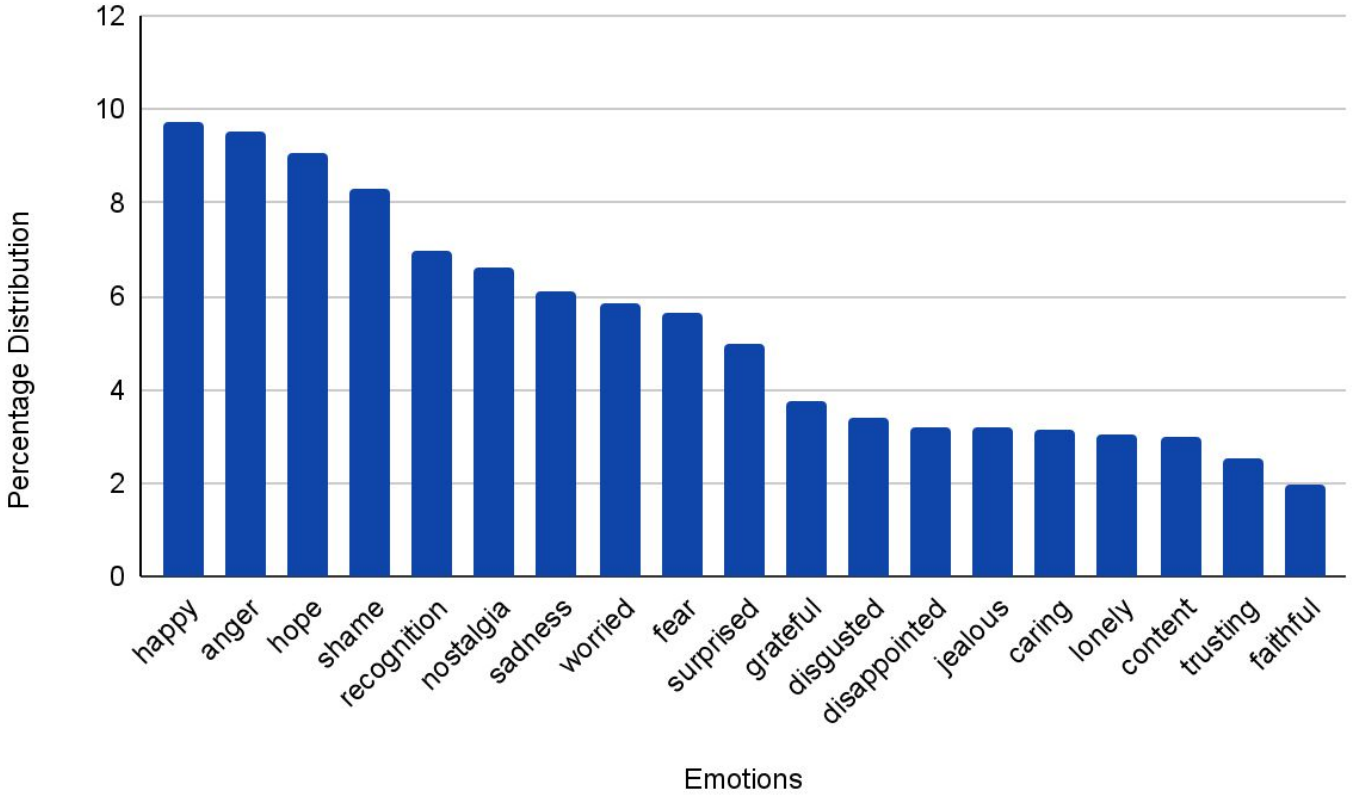


Source:
https://www.researchgate.net/figure/BERT-base-uncased-model-architecture-which-comprises-12-transformer-block-layers-each_fig2_374608193

Emotions Mapping - BERT



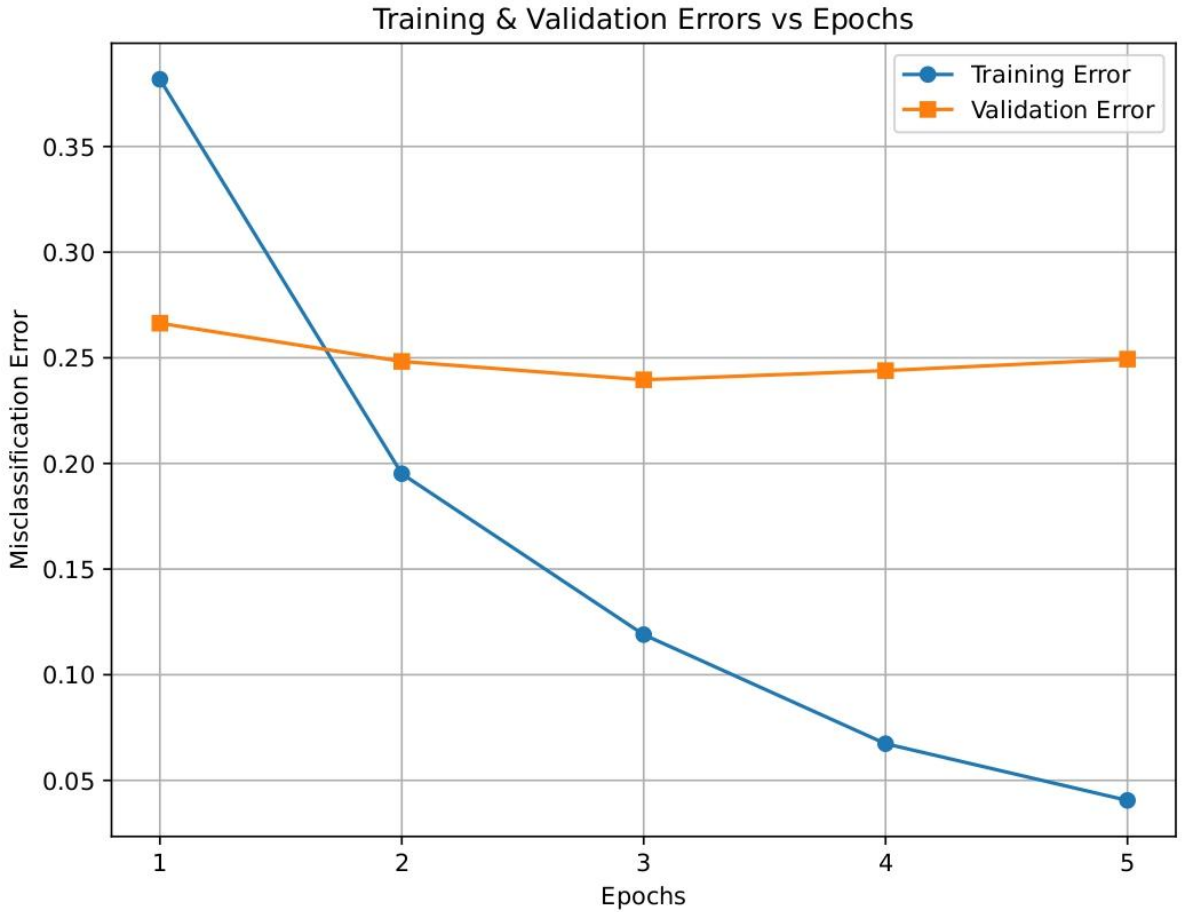
Emotions Mapping - BERT



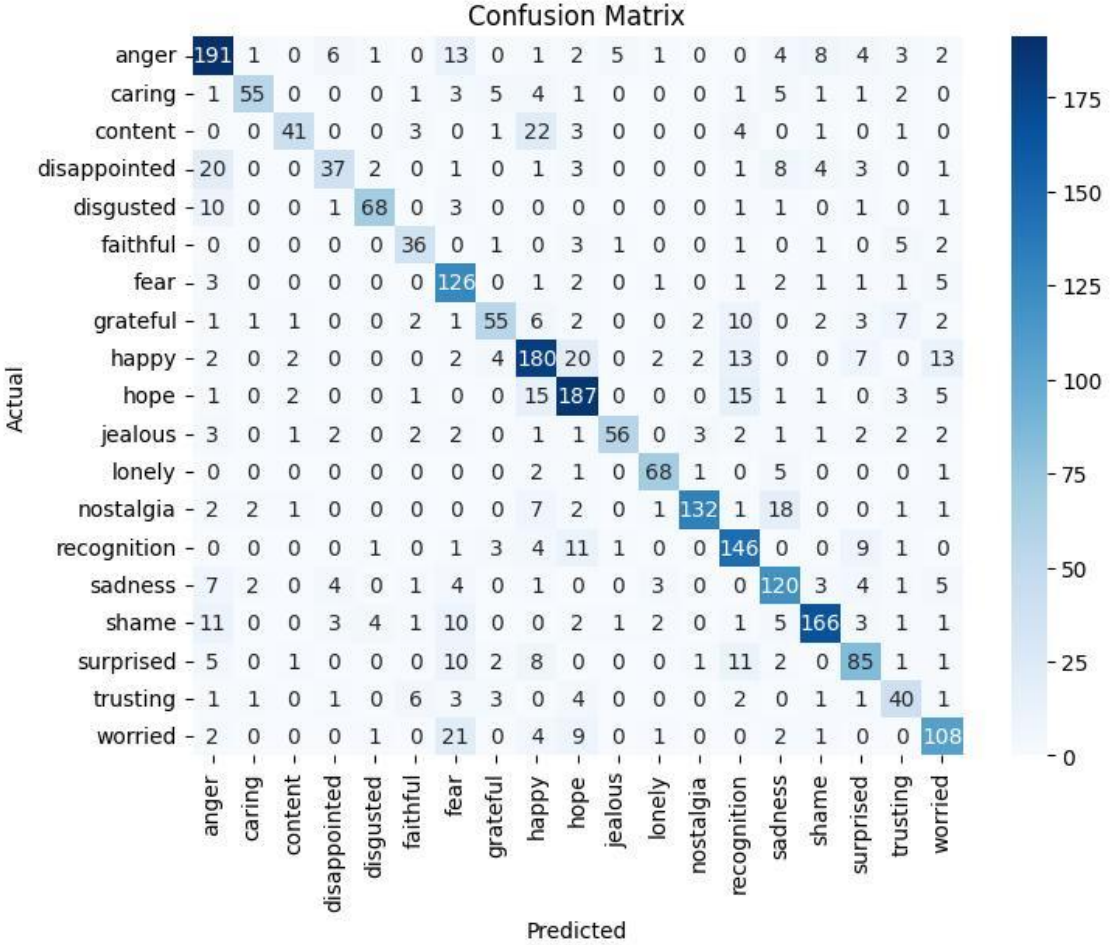
BERT - Model Training

Component	Purpose
AdamW Optimizer	Optimizes BERT while reducing weight decay bias.
CrossEntropyLoss	Computes classification loss.
Mixed Precision (FP16)	Speeds up training & reduces memory usage.
Gradient Accumulation	Allows large batch training on limited memory.
Cosine Annealing LR Scheduler	Smoothly adjusts learning rate to prevent sharp changes.
Validation	Evaluates model performance after each epoch.

Results - Training vs Validation Error



Results - Test Set



Results - Test Set

Emotion	precision	recall	f1-score	support
anger	0.73	0.79	0.76	242
caring	0.89	0.69	0.77	80
content	0.84	0.54	0.66	76
disappointed	0.69	0.46	0.55	81
disgusted	0.88	0.79	0.83	86
faithful	0.68	0.72	0.70	50
fear	0.63	0.88	0.73	144
grateful	0.74	0.58	0.65	95
happy	0.70	0.73	0.71	247
hope	0.74	0.81	0.77	231
jealous	0.88	0.69	0.77	81
lonely	0.86	0.87	0.87	78
nostalgia	0.94	0.79	0.85	168
recognition	0.70	0.82	0.75	177
sadness	0.69	0.77	0.73	155
shame	0.87	0.79	0.83	211
surprised	0.69	0.67	0.68	127
trusting	0.58	0.63	0.60	64
worried	0.72	0.72	0.72	149
accuracy	**0.75**	**0.75**	**0.75**	**0.75**
weighted avg	0.76	0.75	0.75	2542

Results - Test Set

Example 1:
Conversation: I got attacked by a swarm of bees a few months back. Needless to say it hurt. How awful, besides painful, it must have been **scary**! It was, they we're absolutely **furious** with me for some reason. Were you messing with them, maybe? or taking their honey?
True Label: anger
Predicted Label: fear

Example 2:
Conversation: I have to rely on my prayers and my belief in a high power to get me through not having a job and no money. Things work out. It doesn't seem like it but they do. I remind myself all the time that it will work out and all will be good again. It will. I am sure of it.
True Label: faithful
Predicted Label: trusting

Example 3:
Conversation: My husband was teaching me how to make a steak. The flame was so warm! That sounds like a fun thing to do as a couple. It was but I was so sad that I was so **scared** Why were you so **scared**?
True Label: worried
Predicted Label: fear

Conclusion

Aspect	Traditional ML (Hierarchical Model)	Deep Learning (BERT)
Performance	Lower accuracy, F1-score	Higher accuracy, better F1-score
Feature Extraction	Requires manual feature engineering (TF-IDF)	Automatically learns from raw data
Computation	Low (works on CPU)	High (needs GPU)
Speed	Faster training	Slower training
Use case scalability	Suitable for resource-limited environments & simple tasks	Best for complex NLP where accuracy matters more than speed

THANK YOU!